

Project Summary – HealthConnect No-Show Prediction

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Batch: D

1. What I planned to accomplish in Week 5

I planned to move from my Week 4 plan into implementation: prepare the HealthConnect appointment data for modelling, engineer meaningful features, define a train/test strategy, train a baseline classification model, and evaluate it.

2. What I actually completed

I built the full preparation pipeline (missing-value imputation, duplicate checks, encoding of categorical variables), engineered behavioural and scheduling features, split the data 80/20 with stratification (4,000 train / 1,000 test), and trained a logistic regression baseline plus a dummy classifier for comparison. I evaluated both with accuracy, precision, recall, F1, confusion matrices and ROC-AUC.

3. Key findings / development outcomes

My logistic regression reached 46.0% accuracy and 0.628 ROC-AUC, modestly beating the dummy baseline (44.9% / 0.499). The model catches only ~49% of actual no-shows, and the Cancelled class is unpredictable ($F1 = 0.12$). Age-group and day-of-week features dominate the linear model's coefficients.

4. Major challenges encountered

The rare Cancelled class (5.2%) made the 3-class formulation unstable; the linear model

showed weak signal; and rare one-hot categories introduced overfitting risk.

5. Important decisions made and why

I kept median imputation and one-hot encoding, used a stratified split to preserve class balance, and deliberately tested a 3-class model to validate my Week 4 decision to exclude cancellations. The poor Cancelled performance confirmed that decision.

6. Changes to my Week 4 approach

My Week 4 approach proposed a binary no-show model; I temporarily tested a 3-class version in Week 5 to validate that choice. The results confirmed the binary formulation, so I will return to it in Week 6. I also added tree-based models to my plan because the linear baseline under-used my behavioural features.

7. Cross-track collaboration completed

I collaborated with the Data Analytics track: I used their KPI evidence on reminder and lead-time effects to prioritise features, and I shared my baseline results so their dashboard metrics align with model outputs.

8. Remaining work

Retrain with the binary formulation, test tree-based models, apply class balancing, tune hyperparameters and the decision threshold, and re-evaluate with PR curves.

9. Proposed focus for Week 6

Develop and evaluate improved classification models (Random Forest / Gradient Boosting) on the binary no-show task, with threshold tuning aimed at maximising no-show recall, to produce a model ready for the HealthConnect solution.